Can You Hear the Difference?

YouTube Audience Reception of AI vs. Human Narration

CIS Data Mining · Spring 2026 · Research Study

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Abstract

AI voice narrated tools have become an almost default production setting on YouTube over the past 18 months. This study examines how audiences engage with, react to, and discuss AI-narrated videos compared to human-narrated content. Using 644 videos across 25 channels and 39,600 comments scraped via the YouTube Data API v3, we applied Mann-Whitney U tests, VADER sentiment analysis, KMeans clustering with UMAP projection, and keyword co-occurrence analysis. Standard engagement metrics do not consistently separate the two groups. Statistically significant differences were found in comment rate ($p = 0.008$), mean sentiment ($p = 0.035$), and authenticity-flag rate ($p = 0.047$), though only comment rate survives Bonferroni correction ($\alpha_{\text{adj}} = 0.01$). A substantially weaker sentiment-to-engagement coupling is observed on AI-narrated content (Spearman $\rho = 0.185$ vs. $\rho = 0.471$ for human-narrated). We conclude that AI narration does not incur a large engagement penalty in 2026, but it measurably shifts comment section behavior.

Keywords: AI narration, YouTube engagement, VADER sentiment, authenticity, parasocial interaction, KMeans clustering, Bonferroni correction

Table 1: Dataset at a Glance

Metric	Value	Notes
Total Videos	644	358 human · 286 AI
Channels	25	Hand-audited narration labels
Total Comments	39,600	Top-level, 2026 only
Authenticity-Flagged Comments	1,843	Used for H3 analysis

1. Motivation & Research Questions

1.1 The Production Shift

Over the last 18 months, AI voice tools—including ElevenLabs, OpenAI TTS, NotebookLM, and WaveNet—have become default production tools on YouTube. Top-10 lists, facts channels, finance explainers, and even documentary content now regularly feature narration that is largely indistinguishable from a human voice.

1.2 Research Questions & Hypotheses

The core question driving this study: *when controlling for topic and release window, does a YouTube audience engage with, react to, and discuss AI-narrated videos differently than human-narrated ones?*

Three hypotheses were tested:

- **H1 — Engagement:** Like-to-view and comment rates differ between groups.
- **H2 — Sentiment:** The tone of comment sections differs between groups.
- **H3 — Authenticity:** Discussion of voice provenance is more frequent on AI-narrated videos.

2. Related Work

2.1 Prior Literature

Jakesch, Hancock & Naaman (2023, PNAS) demonstrated that human heuristics for AI-generated language are unreliable—in controlled experiments, people cannot consistently distinguish AI-generated text from human writing.

Köbis & Mossink (2021, *Computers in Human Behavior*) found that readers failed to differentiate AI-generated poetry from human-written poetry—a result particularly relevant to the parasocial dimension of voice content.

Horton & Wohl (1956, *Psychiatry*) established the foundational account of parasocial interaction: the one-sided relationships audiences form with media personalities, providing the theoretical backdrop for examining audience responses to synthetic voices.

Table 2: Prior Work vs. This Study

Dimension	Prior Work	This Study
Medium	AI-generated text	Narration, where parasocial stakes are higher
Setting	Lab-based forced-choice detection tasks	Observational YouTube engagement in the wild
Measurement	Explicit ratings of authenticity	Actual likes, comments, and sentiment behavior

2.2 Our Contribution

3. Data Collection

3.1 Source & Instrumentation

Data was acquired via the YouTube Data API v3 using the `channels.list`, `playlistItems.list`, `videos.list`, and `commentThreads.list` endpoints. Filters applied: published in 2026, top-level comments only, ordered by YouTube relevance, minimum 5 words per comment.

3.2 Preprocessing Pipeline

- NFKC normalization → lowercase → strip URLs & HTML entities
- Emoji demojization (`demjize`)
- Filter comments under 5 words
- Compute like/view and comment rates per video

4. Methodology

4.1 Pipeline

The project was divided across four team members, each owning a distinct stage of the pipeline. All results are fully reproducible from a single CLI command (`python main.py --all`) with seed 42 applied throughout.

5. Results

5.1 Inferential Statistics (H1, H2 & H3)

Mann-Whitney U tests were conducted on all 659 videos ($n_{\text{human}} = 372$, $n_{\text{AI}} = 287$). Three of five metrics reached nominal significance at $\alpha = 0.05$; only comment rate survives Bonferroni

Table 3: Team Pipeline

Member	**Role**	**Tools & Methods**
Tiago	Data Acquisition	`google-api-python-client` · Exponential backoff · Per-channel cap 50 · Per-video cap 100
Carlos	Linguistic Analysis	TF-IDF (top 50 terms) · Bigrams & trigrams · Keyword co-occurrence matrix · Authenticity flag list
Henrique	Sentiment & Clustering	MiniLM-L6-v2 (384-d) · KMeans + silhouette tuning · UMAP 2D projection · VADER compound · Logistic Regression, Random Forest & XGBoost classifiers
Natacha	Statistical Testing	Mann-Whitney U (2-sided) · Rank-biserial effect size r · Spearman ρ within groups · $\alpha = 0.05$, Bonferroni correction ($\alpha_{\text{adj}} = 0.01$)

correction ($\alpha_{\text{adj}} = 0.05/5 = 0.01$). Effect sizes are small throughout ($|r| \leq 0.12$), indicating that while some differences are detectable, their practical magnitude is limited.

Table 4: Mann-Whitney U Test Results ($n_{\text{human}} = 372$, $n_{\text{AI}} = 287$). $^{*}p < 0.05$; $^{**}p < 0.01$ (survives Bonferroni).

Metric	**U**	$\boldsymbol{p}$	$\boldsymbol{r}$	$\mathbf{Med_H}$	$\mathbf{Med_{AI}}$	**Sig.**
Like-to-view ratio	52,946	0.857	+0.008	0.028	0.031	n.s.
Comment rate	59,850	0.008	−0.121	0.0020	0.0019	**
Mean sentiment (VADER)	48,276	0.035	+0.096	0.127	0.148	*
% Negative comments	55,159	0.463	−0.033	0.223	0.234	n.s.
Authenticity-flag rate	58,142	0.047	−0.089	0.030	0.021	*

*Nominal only; does not survive Bonferroni correction at $\alpha_{\text{adj}} = 0.01$.

5.2 Sentiment Distributions (VADER)

VADER compound scores were computed for all 39,600 comments. Both groups sit around a mean compound of approximately +0.14. The zero-peak reflects a large neutral mass of short reactions such as *"lol"* or *"thanks."*

Human and AI kernel density estimates overlap almost completely across the full $[-1, +1]$ range with no distinguishable mode shift. The mean sentiment difference, while nominally significant ($p = 0.035$), does not survive multiple-comparison correction and carries a small effect ($r = 0.096$).

5.3 Sentiment-to-Engagement Coupling

A more revealing finding emerges when examining within-group Spearman correlations between mean sentiment and the like-to-view ratio:

Table 5: Within-Group Spearman Correlations: mean sentiment $\sim$ like-to-view ratio.

Group	Spearman ρ	p-value	n	Interpretation
Human-Narrated	+0.471	< 0.001	372	Strong affect-to-action coupling
AI-Narrated	+0.185	0.002	287	Weak coupling — 2.5× lower

When viewers feel positively about a human-narrated video, they click like in proportion ($\rho = 0.471$). On AI-narrated videos, that affect-to-action link is more than twice as weak ($\rho = 0.185$), despite both correlations being individually significant. This decoupling is the study's most practically significant finding and is robust to multiple comparisons since it was a pre-specified within-group analysis.

5.4 Comment Clustering

KMeans ($k = 5$) clustering was applied to MiniLM-L6-v2 embeddings (384-dimensional) of all 39,600 comments, with a 2D UMAP projection for visualization. AI and human comment markers are fully interleaved across all five clusters—narration type does not segregate semantic space.

Table 6: KMeans Comment Clusters ($k = 5$, seed 42).

Cluster	Top Keywords	Human	AI
Generic opinion	like, just, people, don, make	7,980	3,818
Praise of the video	video, love, watching	5,500	5,405
Emoji-heavy fan reactions	blackpink, fan, love	3,714	1,647
News & geopolitics	trump, iran, war, israel	4,788	2,181
Gaming & reviews	game, games, really	984	3,583

5.5 Supervised Sentiment Classifiers

As a secondary analysis, Henrique trained three supervised binary classifiers (positive vs. negative) on TF-IDF features, using weak labels derived from VADER compound scores (> 0.05 = positive, < -0.05 = negative). The train/test split was 80/20 stratified ($n_{\text{train}} =$ 24,188, $n_{\text{test}} = 6{,}047$).

Logistic Regression achieved the highest F1 (0.881) and ROC-AUC (0.922). The top positive-sentiment terms by coefficient weight were *love*, *like*, *great*, *best*, and *good*; the strongest

Table 7: Supervised Sentiment Classifier Performance.

Model	Acc.	Prec.	Recall	F1	ROC-AUC
Logistic Regression	0.849	0.912	0.853	**0.881**	**0.922**
Random Forest	0.837	0.863	0.893	0.878	0.903
XGBoost	0.793	0.776	0.963	0.860	0.873

negative predictors were *war*, *hell*, *bad*, *crazy*, and *dead*. These classifiers served as a validation check on VADER's coverage, not as primary analysis instruments.

5.6 Authenticity Co-Occurrence Network

Within the 1,843 authenticity-flagged comments, keyword co-occurrence analysis reveals that discourse clusters tightly around direct human/AI comparisons.

Table 8: Top Keyword Co-Occurrence Pairs in Authenticity-Flagged Comments.

Rank	Keyword Pair	Count	Frame Type
1	`ai` × `real`	227	Direct comparison framing
2	`ai` × `human`	96	Benchmark framing
3	`ai` × `voice`	63	The accusation
4	`human` × `real`	38	Praise framing
5	`ai` × `fake`	30	Negative framing
6	`sounds like` × `ai`	26	Diagnostic phrasing

Marginal keyword counts: `ai` (913), `real` (736), `voice` (271), `human` (239), `sounds like` (152), `fake` (110), `robot` (57).

6. Discussion

6.1 Interpretation of Results

The null results on two of five engagement metrics (like-to-view ratio and % negative comments) are meaningful. They suggest that, in 2026, AI narration has been sufficiently normalized that most viewers engage on the same terms regardless of voice type.

The comment rate difference ($p = 0.008$, $r = -0.121$) is the only finding robust to Bonferroni correction, indicating that human-narrated videos attract slightly higher comment rates at the median (0.0020 vs. 0.0019). The effect is statistically reliable but practically small.

The sentiment-engagement decoupling ($\rho = 0.471$ vs. 0.185) is the study's most consequential finding. Viewers on AI-narrated channels convert positive affect into engagement action at a substantially lower rate, suggesting a partial breakdown of the parasocial loop that ordinarily connects feeling good about content with signaling approval through likes.

6.2 Threats to Validity

- **Genre confound:** Human and AI strata do not fully overlap in genre. Every narration contrast is therefore also a genre contrast.
- **Channel-level labels:** Narration type is tagged at the channel level from a short audio sample. Channels that mix AI and human narration are assigned to a single category.
- **Comment sampling:** Only top-level comments ordered by YouTube relevance (not random) are included.
- **Multiple comparisons:** Three metrics reach $p < 0.05$; only one survives Bonferroni correction at $\alpha_{\text{adj}} = 0.01$. Mean sentiment and authenticity-flag results should be interpreted with caution.
- **Keyword matching:** The authenticity flag uses substring matching, which may over-flag comments containing partial matches. A word-boundary regex is recommended in future iterations.

7. Conclusions

This study examined 644 YouTube videos, 25 channels, and 39,600 comments to determine whether audiences engage differently with AI-narrated versus human-narrated content in 2026. Four key takeaways emerge:

1. **Most engagement metrics do not separate the groups.** Like-to-view ratio and % negative comments show no significant difference.
2. **Comment rate is reliably higher on human-narrated videos.** $p = 0.008$, $r = -0.121$; the only result surviving Bonferroni correction.
3. **Authenticity talk is nominally more common on AI videos.** $p = 0.047$, though this does not survive multiple-comparison correction and may be inflated by imprecise keyword matching.
4. **Sentiment decouples from engagement on AI content.** $\rho = 0.471$ (human) vs. $\rho = 0.185$ (AI). The loop that converts positive affect into clicking like is substantially weakened on AI-narrated channels.

7.1 Future Work

- Genre-stratified replication to isolate narration effects from genre effects
- Word-boundary regex for authenticity keyword matching to reduce false positives
- Reply-thread inclusion for fuller authenticity discourse analysis
- Multilingual sentiment analysis and embeddings for non-English content
- Longitudinal extension (2022–2026) to trace when authenticity talk emerged
- Controlled re-narration experiment for causal identification

References

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A. Team Contributions

Table 9: Member Contributions

Member	Contributions
Tiago	YouTube Data API v3 integration, exponential backoff implementation, channel corpus construction, per-channel and per-video capping logic, raw CSV output pipeline.
Carlos	TF-IDF term extraction (top 50), bigram and trigram analysis (top 20), keyword co-occurrence matrix construction, authenticity flag list design and application.
Henrique	MiniLM-L6-v2 comment embeddings, KMeans clustering with silhouette tuning, UMAP 2D projection, VADER compound sentiment scoring across all 39,600 comments, supervised sentiment classifiers (Logistic Regression, Random Forest, XGBoost).
Natacha	Mann-Whitney U test design and execution, rank-biserial effect size computation, Spearman ρ within-group correlations, Bonferroni correction ($\alpha_{\text{adj}} = 0.01$) with per-metric significance flags, defensive data-quality guard on authenticity flag rate, results narration.

B. Source Code & Reproducibility

The complete source code and analysis scripts used to generate the findings in this report are open-source and available in our public GitHub repository:

`https://github.com/Gamonal6/social_media_mining`